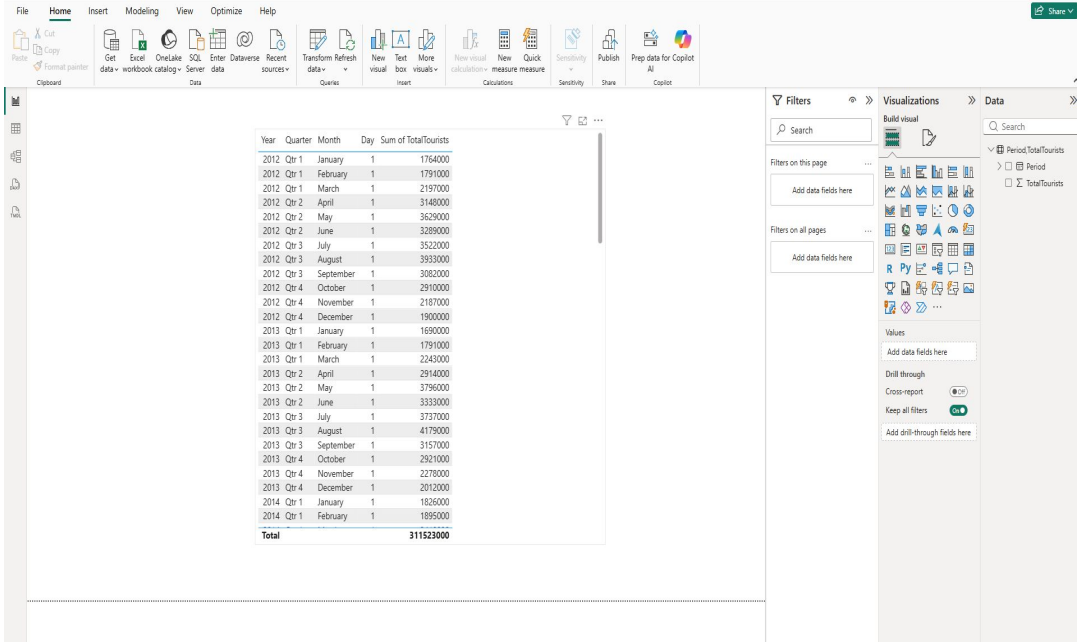
# Forecasting Time Series in Power BI

# Step 1: Download the Sample Dataset

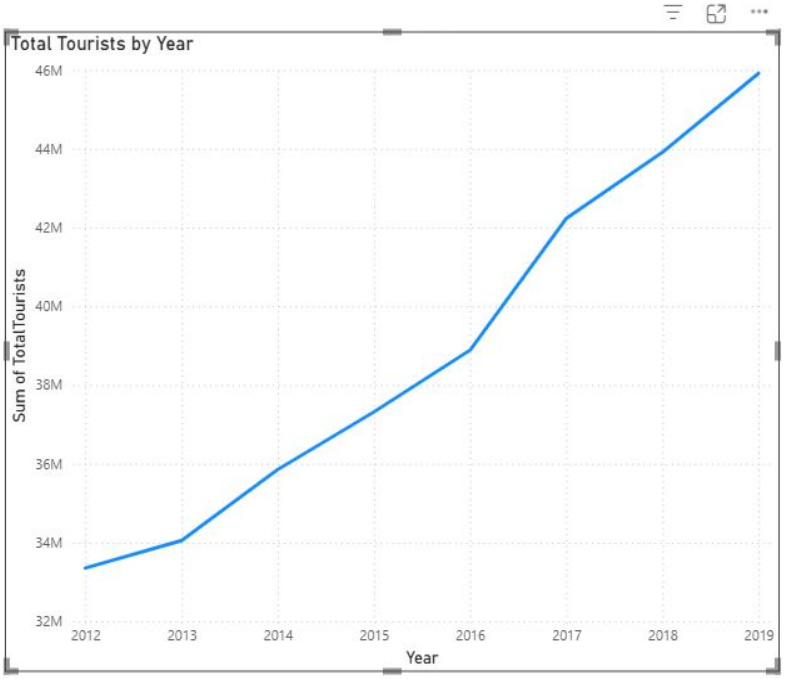
```
Period,Total tourists
1/1/2012,1764000
2/1/2012,1791000
3/1/2012,2197000
4/1/2012,3148000
5/1/2012,3629000
6/1/2012,3289000
7/1/2012,3522000
8/1/2012,3933000
9/1/2012,3082000
10/1/2012,2910000
11/1/2012,2187000
12/1/2012,1900000
1/1/2013,1690000
2/1/2013,1791000
3/1/2013,2243000
4/1/2013,2914000
5/1/2013,3796000
6/1/2013,3333000
7/1/2013,3737000
8/1/2013,4179000
9/1/2013,3157000
10/1/2013,2921000
11/1/2013,2278000
12/1/2013,2012000
1/1/2014,1826000
2/1/2014,1895000
3/1/2014,2410000
4/1/2014,3162000
5/1/2014,3626000
6/1/2014,3868000
7/1/2014,3700000
8/1/2014,4281000
9/1/2014,3368000
10/1/2014,3270000
11/1/2014,2331000
12/1/2014,2118000
1/1/2015,1889000
2/1/2015,2106000
3/1/2015,2373000
4/1/2015,3272000
5/1/2015,4036000
6/1/2015,3617000
7/1/2015,4079000
8/1/2015,4449000
9/1/2015,3316000
10/1/2015,3401000
11/1/2015,2522000
12/1/2015,2258000
1/1/2016,2048000
2/1/2016,2324000
3/1/2016,2704000
4/1/2016,3278000
5/1/2016,4165000
6/1/2016,3526000
7/1/2016,4085000
8/1/2016,4527000
9/1/2016,3675000
10/1/2016,3615000
11/1/2016,2571000
```

# Step 2: Import Data into Power BI




| Year | Quarter | Month | Day | Sum of TotalTourists |
|---|---|---|---|---|
| 2012 | Qtr 1 | January | 1 | 1764000 |
| 2012 | Qtr 1 | February | 1 | 1791000 |
| 2012 | Qtr 1 | March | 1 | 2197000 |
| 2012 | Qtr 2 | April | 1 | 3148000 |
| 2012 | Qtr 2 | May | 1 | 3629000 |
| 2012 | Qtr 2 | June | 1 | 3289000 |
| 2012 | Qtr 3 | July | 1 | 3522000 |
| 2012 | Qtr 3 | August | 1 | 3933000 |
| 2012 | Qtr 3 | September | 1 | 3082000 |
| 2012 | Qtr 4 | October | 1 | 2910000 |
| 2012 | Qtr 4 | November | 1 | 2187000 |
| 2012 | Qtr 4 | December | 1 | 1900000 |
| 2013 | Qtr 1 | January | 1 | 1690000 |
| 2013 | Qtr 1 | February | 1 | 1791000 |
| 2013 | Qtr 1 | March | 1 | 2243000 |
| 2013 | Qtr 2 | April | 1 | 2914000 |
| 2013 | Qtr 2 | May | 1 | 3796000 |
| 2013 | Qtr 2 | June | 1 | 3333000 |
| 2013 | Qtr 3 | July | 1 | 3737000 |
| 2013 | Qtr 3 | August | 1 | 4179000 |
| 2013 | Qtr 3 | September | 1 | 3157000 |
| 2013 | Qtr 4 | October | 1 | 2921000 |
| 2013 | Qtr 4 | November | 1 | 2278000 |
| 2013 | Qtr 4 | December | 1 | 2012000 |
| 2014 | Qtr 1 | January | 1 | 1826000 |
| 2014 | Qtr 1 | February | 1 | 1895000 |
| **Total** | | | | **311523000** |

# Step 3: Create a Line Chart for Yearly Data



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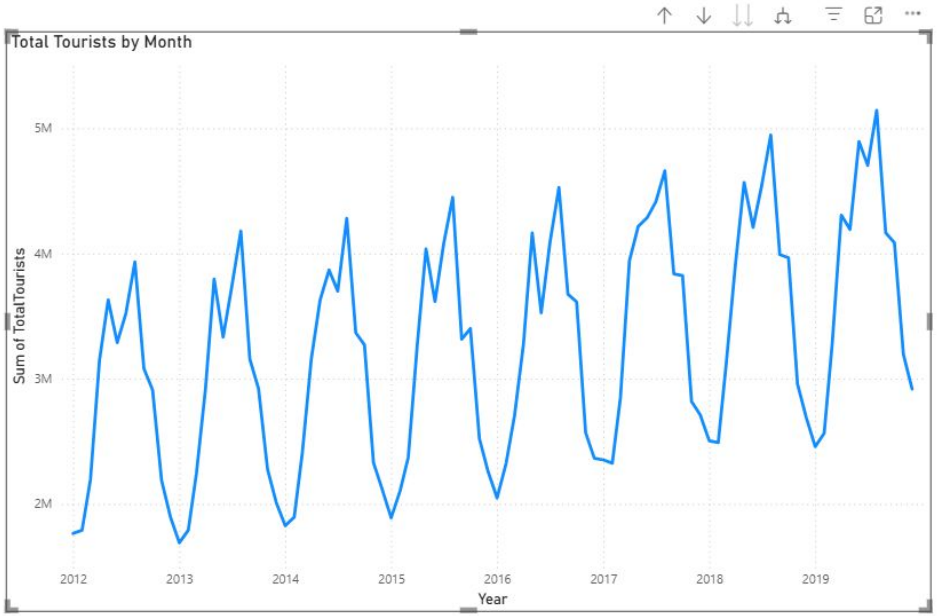
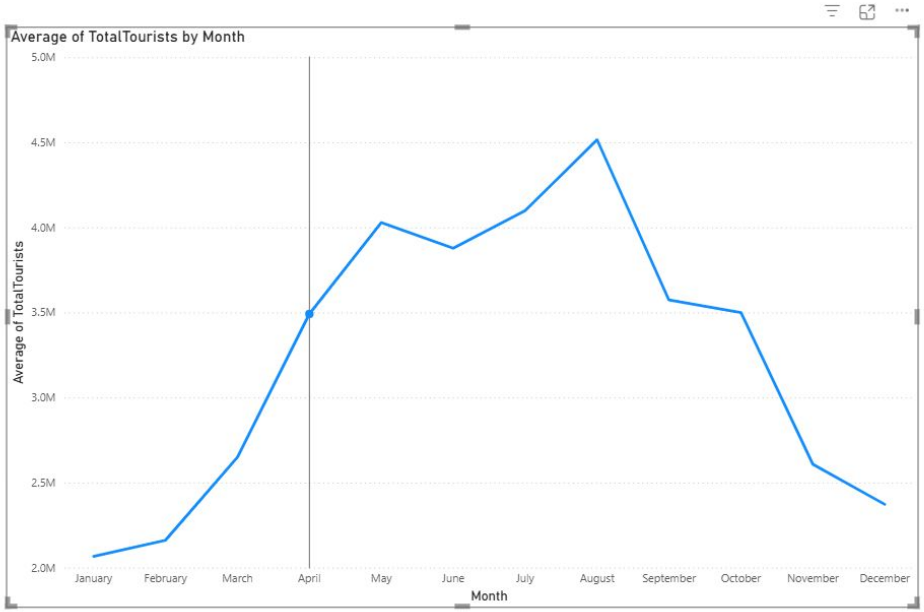
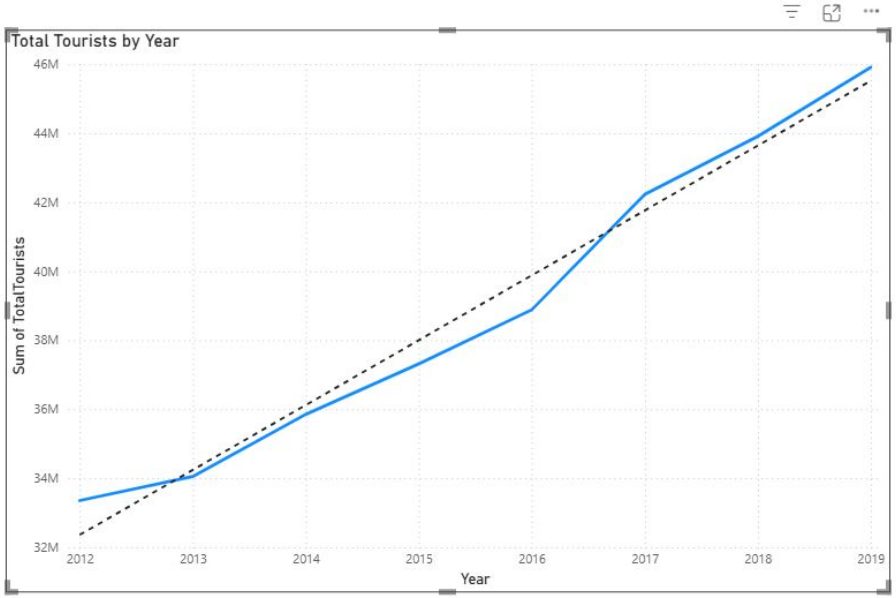


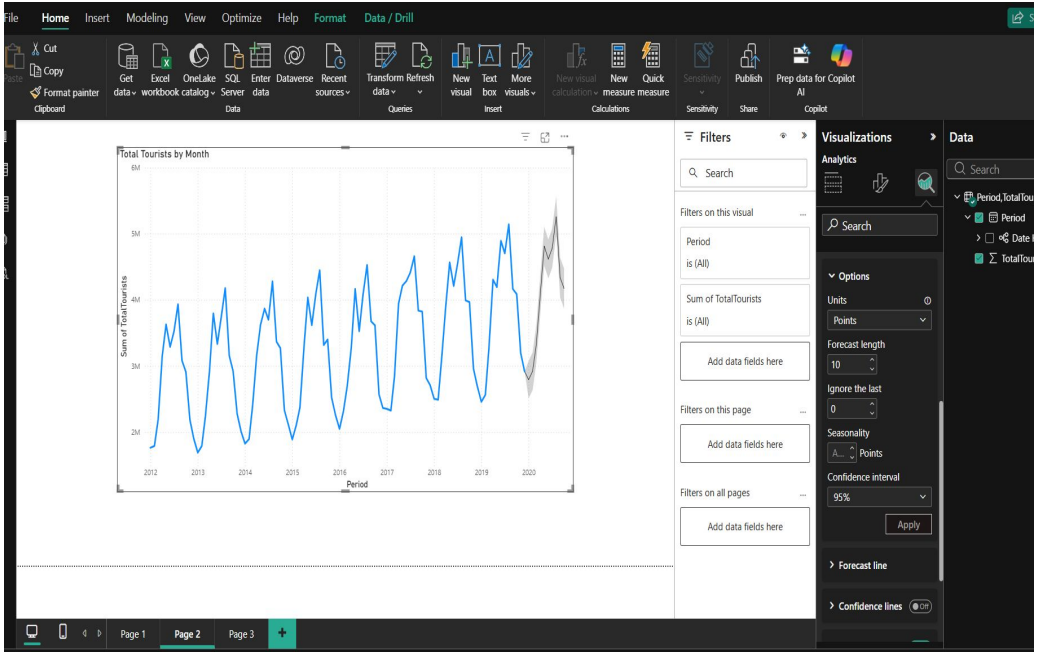
# Step 3: Create a Line Chart for Yearly Data



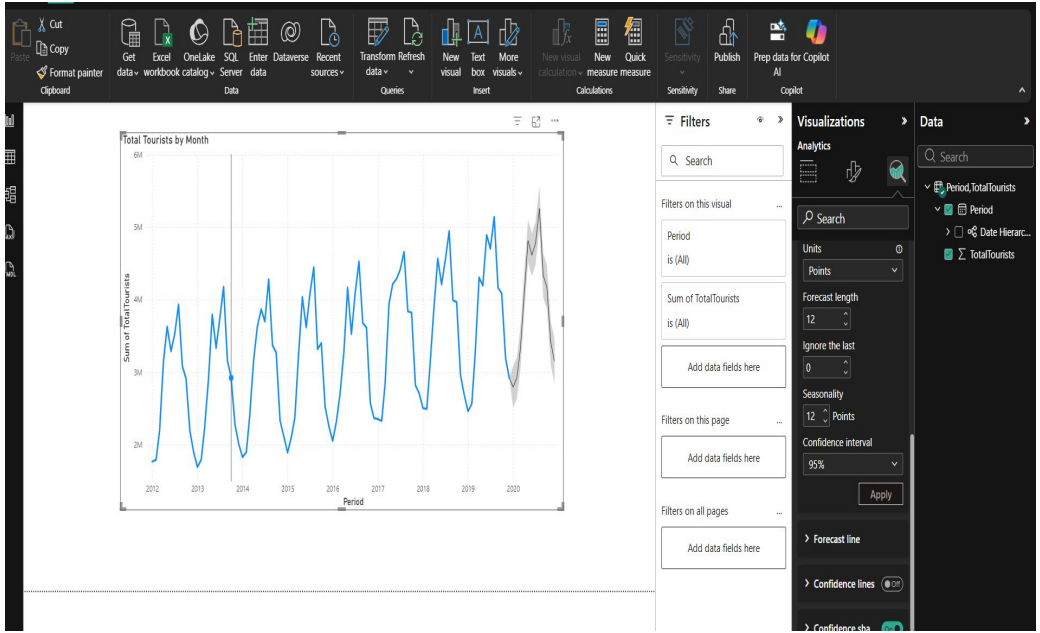
# Step 4: Add a Trend Line (Optional but Recommended)



# Step 5: Add Forecasting



# Step 6: Configure the Forecast



# Step 7: Analyze the Forecast

**1. Forecast for 2020** - The 2020 forecast continues the same wavy seasonal pattern seen from 2012 - 2019. The grey shaded area on the right side of the chart predicts monthly tourist arrivals will keep fluctuating between roughly 3M and 5M, following the same peaks and dips the model learned from 8 years of data.

**2. What the Narrow Confidence Interval Means** - The grey band is relatively narrow, meaning the model is fairly confident in its prediction. This is because the seasonal pattern from 2012 - 2019 was very stable and consistent since the data was so predictable year after year, the model produced a tighter forecast range instead of a wide, uncertain one.

**3. Forecast vs. 2019 Actual Data -** The forecast closely matches the trend seen in 2019. It maintains similar peak values of around 4M - 5M tourists during high seasons and reflects the gradual upward growth observed across all years. This alignment shows the model successfully captured the seasonal behavior of the data.

# Part 2: Enhancement - Forecasting with Imperfect Data

# Step 8: Prepare a New Dataset (or use a modified version)

File Edit View

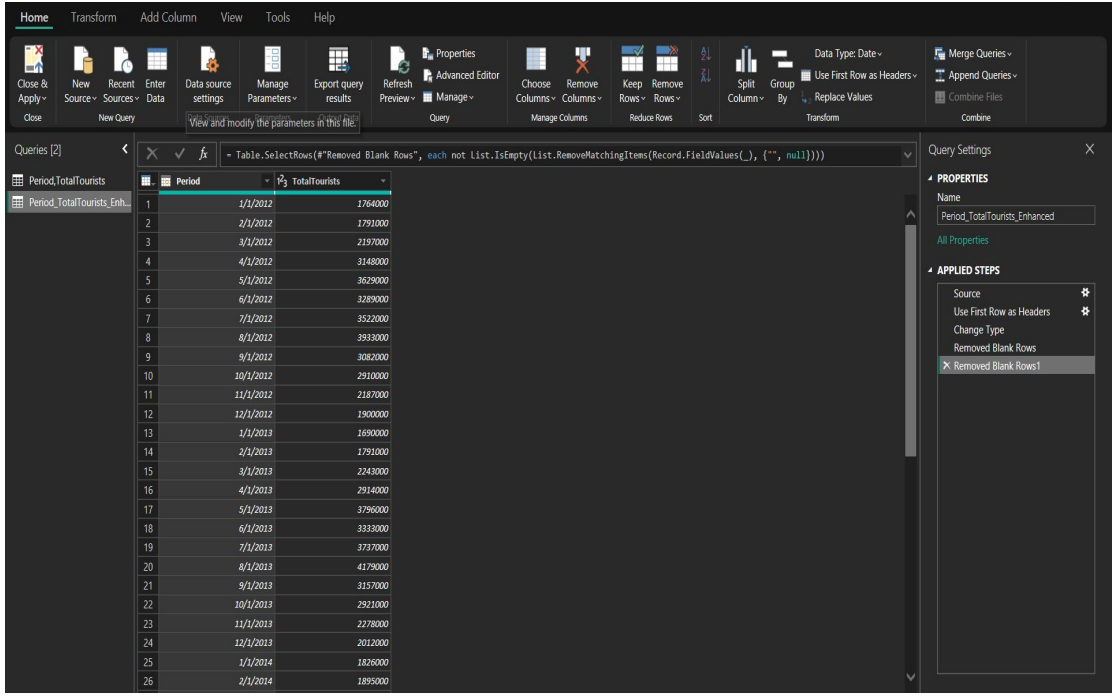
```
2022-01-01,1102000
2022-02-01,1298000
2022-03-01,1876000
2022-04-01,2543000
2022-05-01,3102000
2022-06-01,3467000
2022-07-01,3891000
2022-08-01,4123000
2022-09-01,3544000
2022-10-01,3278000
2022-11-01,2456000
2022-12-01,2198000
2023-01-01,2312000
2023-02-01,2478000
2023-03-01,3156000
2023-04-01,4089000
2023-05-01,4523000
2023-06-01,4812000
2023-07-01,5034000
2023-08-01,5367000
2023-09-01,4478000
2023-10-01,4201000
2023-11-01,3312000
2023-12-01,3089000
2024-01-01,2876000
2024-02-01,2934000
```

Ln 1, Col 1 | 2,777 characters | Plain text | 100% | Unix (LF) | UTF-8

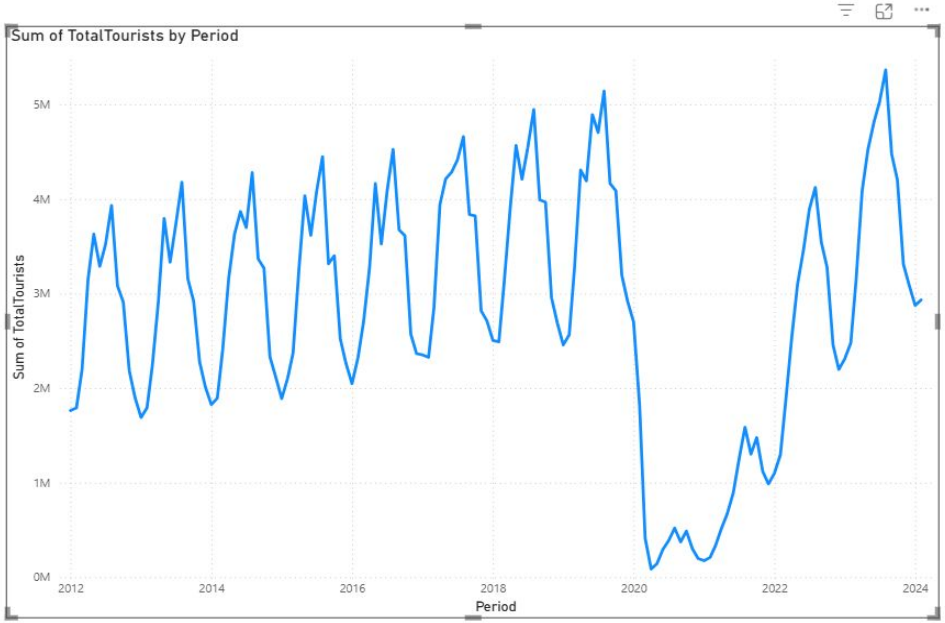
# Step 8: Prepare a New Dataset (or use a modified version)

| Year | Quarter | Month | Day | Sum of TotalTourists |
|---|---|---|---|---|
| 2012 | Qtr 1 | January | 1 | 1764000 |
| 2012 | Qtr 1 | February | 1 | 1791000 |
| 2012 | Qtr 1 | March | 1 | 2197000 |
| 2012 | Qtr 2 | April | 1 | 3148000 |
| 2012 | Qtr 2 | May | 1 | 3629000 |
| 2012 | Qtr 2 | June | 1 | 3289000 |
| 2012 | Qtr 3 | July | 1 | 3522000 |
| 2012 | Qtr 3 | August | 1 | 3933000 |
| 2012 | Qtr 3 | September | 1 | 3082000 |
| 2012 | Qtr 4 | October | 1 | 2910000 |
| 2012 | Qtr 4 | November | 1 | 2187000 |
| 2012 | Qtr 4 | December | 1 | 1900000 |
| 2013 | Qtr 1 | January | 1 | 1690000 |
| 2013 | Qtr 1 | February | 1 | 1791000 |
| 2013 | Qtr 1 | March | 1 | 2243000 |
| 2013 | Qtr 2 | April | 1 | 2914000 |
| 2013 | Qtr 2 | May | 1 | 3796000 |
| 2013 | Qtr 2 | June | 1 | 3333000 |
| 2013 | Qtr 3 | July | 1 | 3737000 |
| 2013 | Qtr 3 | August | 1 | 4179000 |
| 2013 | Qtr 3 | September | 1 | 3157000 |
| 2013 | Qtr 4 | October | 1 | 2921000 |
| 2013 | Qtr 4 | November | 1 | 2278000 |
| 2013 | Qtr 4 | December | 1 | 2012000 |
| 2014 | Qtr 1 | January | 1 | 1826000 |
| 2014 | Qtr 1 | February | 1 | 1895000 |
| 2014 | Qtr 1 | March | 1 | 2410000 |
| 2014 | Qtr 2 | April | 1 | 3162000 |
| **Total** | | | | **415312000** |

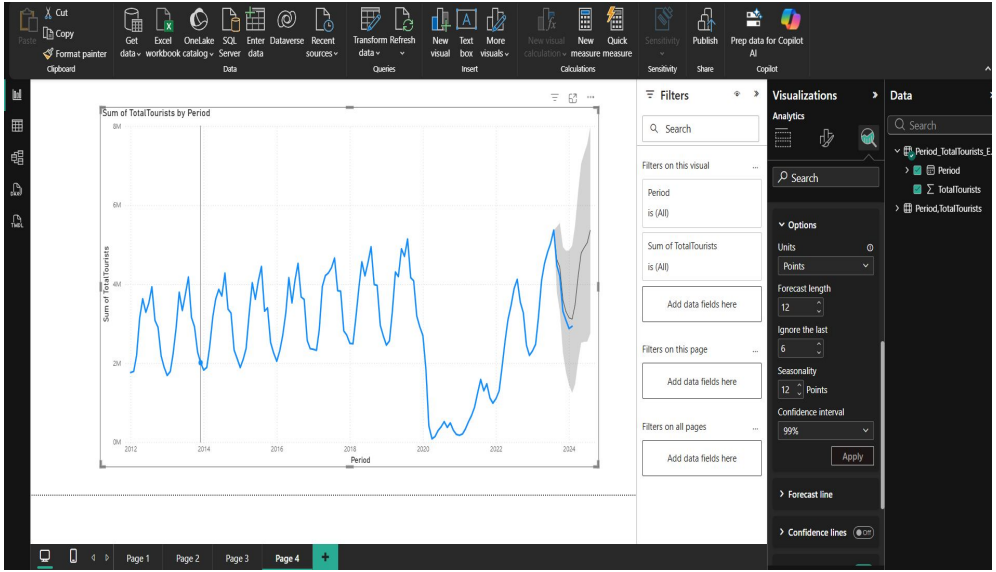
# Step 9: Import and Prepare Your New Dataset



# Step 10: Create a Line Chart for Your Data



# Step 11: Add and Configure the Forecast




What does this tell you about the model's confidence?

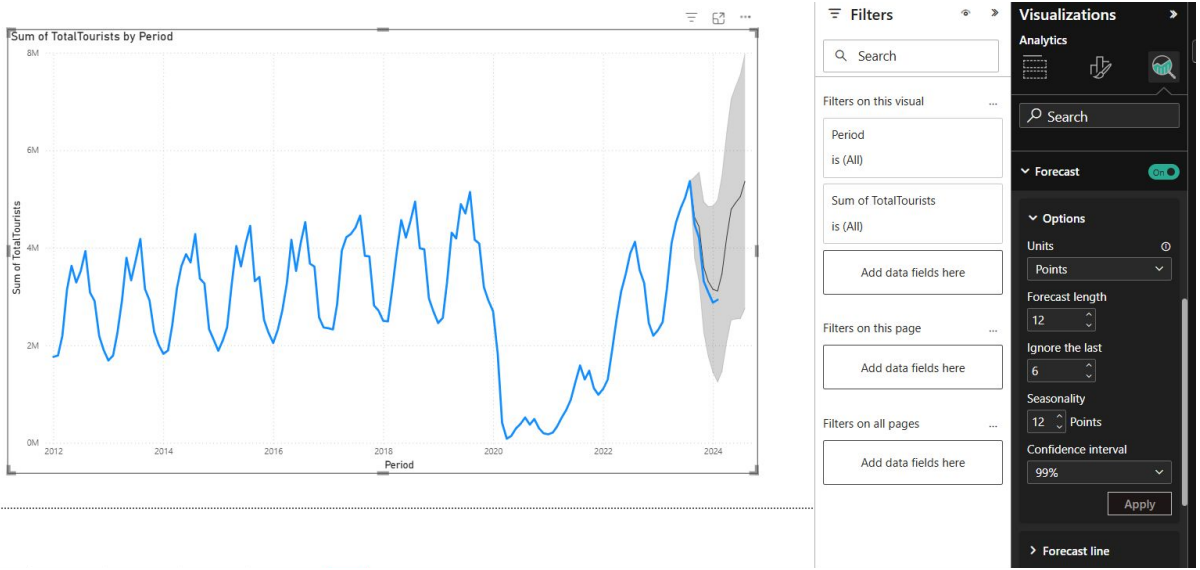
A 99% confidence interval is like asking for a near-guaranteed forecast range. To be that certain, the model has to widen the shaded area it's basically saying, "I'm not sure of the exact number, but I'm almost completely certain it will fall somewhere within this range." The higher the confidence level, the wider the band needs to be.

## Step 12: Analyze and Report Your Findings

1. Describe your dataset: Briefly explain what data you used, its source, and the time period covered.

For this activity, I used an International Tourism dataset that tracks monthly tourist arrivals from January 2012 to December 2019. I then added 50 extra data points extending to February 2024 to simulate real - world conditions, this includes the disruptions caused by the pandemic and the recovery period that followed.

2. Visualize your forecast: Include a screenshot of your Power BI line chart with the forecast applied.



# 3. Interpret the forecast

**Trend and Seasonality** - The data follows a clear 12 - month seasonal pattern with consistent peaks and dips each year. There was a steady upward trend until early 2020, when a sudden massive drop occurred. Recovery began around 2022, with the seasonal pattern returning but with more unpredictability than before.

**The Power BI Forecast** - The model projects the recovery forward using a black line, correctly identifying the annual cycle and predicting that the ups and downs will continue into 2024 and 2025.

**Confidence Interval** - Setting the interval to 99% produced a wide grey band. This means the model can estimate the general direction, but isn't confident about exact numbers largely because the 2020 - 2022 period was so chaotic, so it leaves a large margin for error to be safe.

**Settings Adjustment** - Ignoring the last 6 months allowed me to compare the forecast against actual known data. The model turned out to be slightly more conservative than the real recovery, which was faster than expected. Raising the confidence interval to 99% made the forecast more realistic about uncertainty a 90% interval would have looked too precise for such unpredictable data.

**Handling the Disruption** The model didn't predict the 2020 crash since it was an outlier, but it responded by significantly widening the confidence interval essentially acknowledging that the stable patterns from 2012–2019 may no longer fully apply.

## 4. Conclusion

Power BI's forecasting tool is useful for spotting long-term trends and seasonal patterns in time-series data. It's easy to use, automatically detects recurring cycles, and requires little technical knowledge. The Ignore Last feature also allows for basic backtesting against known data.

That said, its limitations show when dealing with sudden, extreme disruptions like the 2020 tourism crash. The model treats these as statistical outliers rather than major shifts, and while it responds by widening the confidence interval to reflect uncertainty, it doesn't offer the deep customization that dedicated statistical tools provide.

Overall, Power BI forecasting works well for general trend analysis and business planning, but should be used carefully when the data is highly irregular or when major unexpected events are involved.